# Unit 1 — The Nature of Innovation

## Warm-up

- Is every invention an innovation? Is every innovation an invention?
- Name one innovation that changed your daily life without involving any new technology.
- Who should get the credit for an innovation: the inventor, the engineer who scales it, or the company that sells it?

## Reading: Beyond the Light Bulb

Ask most people to picture an innovator and they will conjure up a lone genius in a cluttered workshop, waiting for the moment of illumination. The image is seductive, but it is largely a myth. Thomas Edison did not invent the light bulb; more than twenty inventors had produced incandescent lamps before him. His contribution lay elsewhere: in a carbon filament that lasted long enough to be commercially viable, and, above all, in the generating stations, wiring and meters that turned a laboratory curiosity into a system people would pay for.

This distinction between *invention* and *innovation* sits at the heart of the discipline. An invention is the creation of something new; an innovation is a novelty that has been successfully adopted and generates value, whether economic, social or environmental. Many patented inventions never become innovations because nobody can manufacture them affordably, nobody needs them, or nobody is willing to change their behaviour to use them.

Scholars commonly classify innovation along two dimensions. The first concerns its *object*. Product innovation delivers a new or improved good or service; process innovation changes how things are made or delivered, as when Toyota's production system slashed waste on the assembly line. Business- model innovation rearranges how value is captured — think of software sold as a subscription rather than a one-off licence. Organisational innovation, meanwhile, alters how people work together.

The second dimension concerns *degree*. Incremental innovations refine what already exists: a battery that charges ten per cent faster, a lighter aircraft seat. Radical innovations depart sharply from existing technology and often render established skills obsolete. Between the two lie architectural innovations, which reconfigure known components in novel ways. The smartphone contained few genuinely new parts; its achievement was integration.

It is tempting to treat radical innovation as superior, but the evidence suggests otherwise. Incremental improvements, accumulated over decades, account for the bulk of productivity growth in most industries. The efficiency of solar panels, for instance, has risen steadily through thousands of small refinements rather than a single breakthrough. Firms that neglect incremental work in pursuit of the next revolution frequently squander the advantages they already hold.

Innovation is also far less solitary than legend suggests. Modern engineering challenges — from fusion reactors to autonomous vehicles — require teams that blend mechanical, electrical and software expertise with knowledge of regulation, markets and users. Research on patents shows that team- authored inventions have steadily displaced those of individuals and tend to be cited more often. The "eureka moment", where it exists at all, is typically the visible tip of years of collective, often tedious, effort.

Finally, innovation is not inherently benign. New technologies redistribute power, create winners and losers, and generate side effects their creators did not foresee. The engineer who designs an innovation therefore bears a responsibility not only to make it work, but to ask for whom it works, and at what cost. Those questions will recur throughout this course.

**A. Comprehension 1. True, False or Not Given?**

1. Edison was the first person to produce an incandescent lamp.
2. According to the text, an invention becomes an innovation only once it is adopted and creates value.
3. Toyota's production system is an example of business-model innovation.
4. The smartphone is presented as an architectural innovation.
5. Radical innovations are responsible for most productivity growth.
6. Team-authored patents are more profitable than individual ones.
7. The author believes engineers should consider the social consequences of their work.

**2. Answer the questions in your own words.**

1. Why does the author describe the image of the lone genius as "seductive" but a "myth"?
2. Give three reasons from the text why an invention might fail to become an innovation.
3. What does the example of solar panels illustrate?
4. What does the phrase "the visible tip of years of collective effort" imply about eureka moments?

**3. Inference.** What attitude does the author take towards firms that focus only on radical breakthroughs? Quote the words that reveal it.

**B. VocabularyTask 1 — Matching.** Match each term (1–6) with its definition (a–f).

| Term | Definition |
|---|---|
| 1. incremental innovation | a. a novel arrangement of existing components |
| 2. radical innovation | b. a change in how goods are produced or delivered |
| 3. architectural innovation | c. a small improvement to an existing product |
| 4. process innovation | d. a change in how a company captures value |
| 5. business-model innovation | e. a sharp break with existing technology |
| 6. adoption | f. the acceptance and use of a novelty by users |

**Task 2 — Word formation.** Complete each sentence with the correct form of the word in brackets.

1. The project failed because the design lacked commercial _____. (VIABLE)
2. Planned _____ is a controversial strategy in consumer electronics. (OBSOLETE)
3. The team's greatest strength was its _____ of hardware and software expertise. (INTEGRATE)
4. We need a more _____ approach to testing. (SYSTEM)
5. The new alloy was a genuine _____ in materials science. (BREAK THROUGH)
6. Critics questioned the _____ of the claims made in the press release. (VALID)

**Task 3 — Fill in the blanks.** Use the words from the box. There is one extra word. *squander · render · conjure up · steadily · foresee · blend · displace*

1. Digital cameras did not merely compete with film; they began to _____ it entirely.
2. Good teams _____ technical depth with commercial awareness.
3. Efficiency has risen _____ over the past three decades.
4. No one could _____ the security problems the device would later create.
5. Poor management can _____ even a strong technological lead.
6. A single software update can _____ older hardware unusable.

**Task 4 — Collocations.** Which verb does NOT collocate with the noun?

1. *drive / foster / stifle / make* innovation
2. *file / grant / infringe / do* a patent
3. *gain / lose / hold / take* a competitive advantage
4. *meet / exceed / reach / fulfil* expectations